

MINIMAX ALGORITHM

AIM

To implement the **Minimax Algorithm** for game playing in Python to determine the optimal move for a player.

ALGORITHM

1. Start.
2. Define the terminal node values.
3. If a leaf node is reached, return its value.
4. If it is the Maximizer's turn, choose the maximum value.
5. If it is the Minimizer's turn, choose the minimum value.
6. Repeat until the root node is evaluated.
7. Display the optimal value.
8. Stop.

SAMPLE CODE

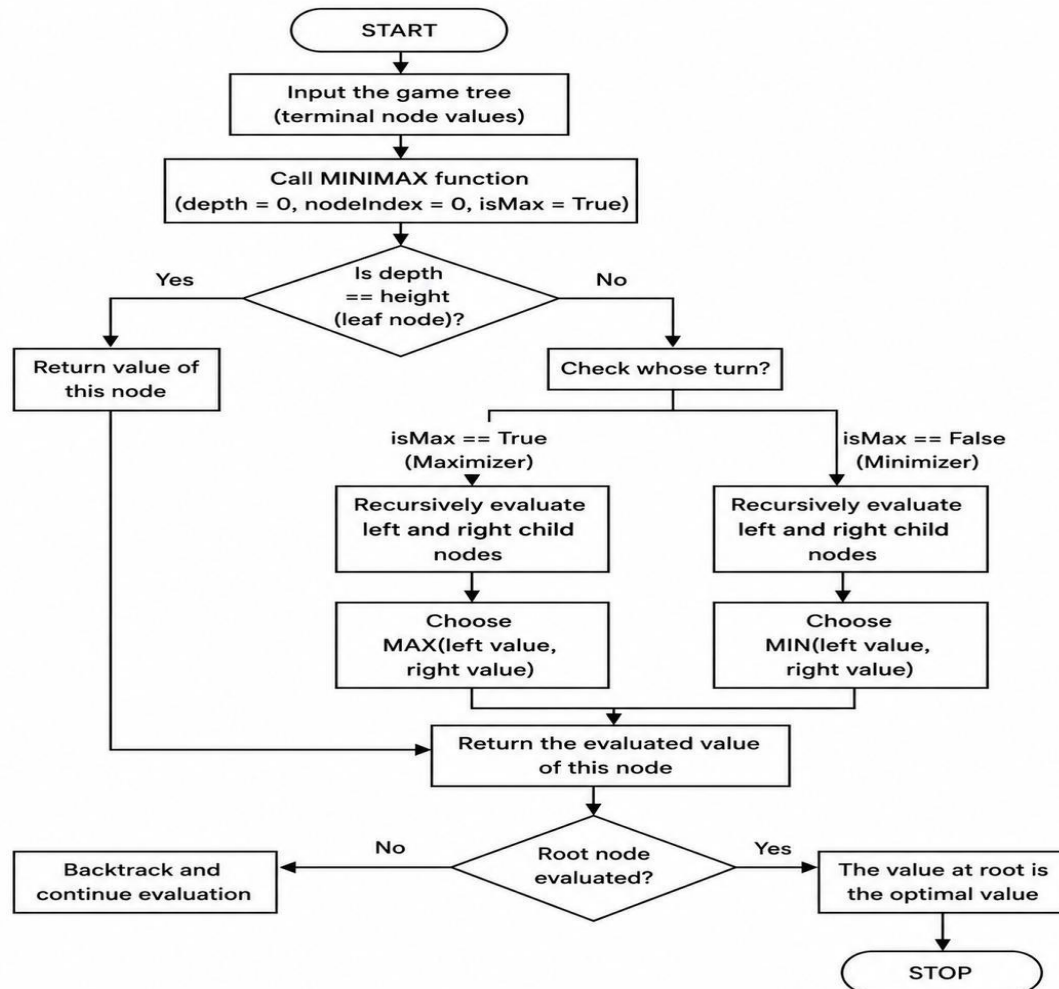
```
# Minimax Algorithm for Gaming
def minimax(depth,
nodeIndex, isMax, values, height):
    if depth ==
height:
        return values[nodeIndex]
    if
isMax:
        return max(
            minimax(depth + 1, nodeIndex * 2, False,
values, height),
            minimax(depth + 1, nodeIndex * 2 + 1,
False, values, height)
        )
    else:
        return min(
            minimax(depth + 1, nodeIndex * 2, True,
values, height),
            minimax(depth + 1, nodeIndex * 2 + 1,
True, values, height)
        )

# Terminal node values
values =
[3, 5, 2, 9, 12, 5, 23, 23]
#
Height of the game tree
height
= 3
```

```
result = minimax(0, 0, True, values, height)
```

```
print("The optimal value is:", result)
```

FLOW CHART



OUTPUT

```
The optimal value is: 12
```

```
Process finished with exit code 0
```